

KARIM TAREK MOHAMED

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SUMMARY

Backend Developer focused on building scalable ASP.NET Core systems with experience in authentication, workflow-driven APIs, integrations, caching, and real-time communication

EDUCATION

Bachelor of Arts (B.A.), Zagazig University — 1st Year *Expected 2029*

Self-Studied: Data Structures, OOP, Database Design, Algorithms, ASP.NET Core, REST APIs, Authentication & Authorization

EXPERIENCE

Freelance Backend Developer — LogiMind Logistics Platform *May 2025 – Present*

- **Implemented shipment lifecycle state management** across 14 logistics workflow states with guarded status transitions, role-restricted commands, hold/resume/cancellation flows, milestone timestamps, and shipment history auditing, improving operational traceability for end-to-end freight operations
- **Developed carrier-based pricing and quotation workflows** with active rate validation, route/container matching, automatic quote calculation, charge aggregation, and shipment snapshot consistency, ensuring accepted quotations preserve agreed carrier, route, pricing, and currency data during execution
- **Engineered transactional rate import workflows** for N8N, carrier APIs, and email-based integrations, implementing idempotent external message tracking, duplicate detection, correlation metadata, unique integration constraints, and per-item import results for reliable carrier pricing ingestion
- **Built 100+ RESTful ASP.NET Core API endpoints** covering authentication, pricing, quotations, shipments, invoices, tracking timelines, document management, customer workflows, and external integrations for a modular freight forwarding platform
- **Implemented secure backend infrastructure** using ASP.NET Core Identity, JWT authentication, hashed refresh token rotation, role-based authorization, email confirmation, phone OTP verification, request rate limiting, and centralized exception handling across admin, staff, customer, and integration workflows
- **Built secure shipment document management workflows** with typed document uploads, ownership-based authorization, extension and file-size validation, ClamAV malware scanning, sanitized storage paths, integration message linking, and soft-delete support for bills of lading, invoices, payment proofs, and customs documents
- **Designed invoice, charge, and payment workflows** with shipment-status-aware business rules, charge-to-invoice linking, subtotal and tax calculation, payment status transitions, cancellation handling, unique invoice numbering, and payer classification across shipper, consignee, and third-party billing scenarios
- **Optimized data access and operational search workflows** using EF Core indexes, pagination, filtering, sorting, eager-loaded aggregate queries, global soft-delete filters, and SQL-backed search across shipments, rates, quotations, customers, routes, carriers, and ports, reducing expensive full-table operational lookups
- **Created validation-heavy backend workflows** using FluentValidation and centralized application rules for shipment tracking dates, invoice inputs, carrier SCAC codes, route uniqueness, customer tax identity, item weights, currencies, and integration payloads, reducing invalid operational data before persistence

PROJECTS

Dental Clinic Management System

- **Built modular ASP.NET Core Web APIs** for dental clinic operations (appointments, billing, **authentication**, ratings, products, cart, orders, payments) across 5+ operational domains with 15+ fully documented REST endpoints and <150ms response times
- **Implemented secure authentication and authorization** using **JWT**, RBAC, refresh tokens, OTP verification, Google sign-in, request sanitization, and rate limiting, protecting patient data and enabling role-based workflows for 3+ user types
- **Implemented Redis caching and query optimization** for high-frequency read operations, reducing repeated database access and improving cached response times from ~300ms to <80ms
- **Documented 15+ REST API endpoints** using FluentValidation, Repository and Unit-of-Work patterns, centralized exception middleware, DTO mapping, and Swagger/OpenAPI, enabling seamless frontend integration

GitHub Repository: [Dental-Clinic-System](#)

UCourses Learning Platform

- **Built scalable ASP.NET Core REST APIs** supporting Admin, Instructor, and Student workflows with role-based authorization across 20+ endpoints and multiple course management modules
- **Integrated Paymob payment gateway** with secure HMAC callback verification and automated enrollment activation workflows, handling secure payment processing and enrollment confirmation flows
- **Implemented real-time notifications and chat workflows** using **SignalR** alongside scheduled background processing with **Hangfire**, enabling low-latency message delivery and automated background task execution
- **Hardened platform security** using **JWT authentication**, refresh token rotation, rate limiting, token blacklisting, email/phone verification, and centralized exception handling across 20+ API endpoints

GitHub Repository: [Courses-Project](#)

TECHNICAL SKILLS

- **Languages:** C#, SQL
- **Frameworks / Libraries:** ASP.NET Core, .NET 8+, SignalR, xUnit
- **Database & ORM:** SQL Server, Entity Framework Core (EF Core)
- **Architecture & Patterns:** Clean Architecture, SOLID, Repository Pattern, Unit of Work, REST API Design, Workflow Design, State Machines
- **Authentication & Security:** JWT, OAuth 2.0, RBAC, Refresh Tokens, HTTPS
- **Caching & Performance:** Redis, Query Optimization, Pagination, Rate Limiting
- **Real-Time & Background Processing:** SignalR, Hangfire
- **Tools & Integrations:** Paymob, SendGrid, Twilio, Google OAuth, n8n, Git, GitHub, Postman, Swagger / OpenAPI